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Supplementary Materials

Materials and Methods

Fitting procedure: For the main analyses, we fitted a separate regularized logistic regression model for each data set via a maximum likelihood estimation. Final model performance was measured through the predictive log-likelihood on hold-out data obtained using a 100-fold cross-validation procedure. In each fold, we split the data into a training set (90%), a validation set (9%), and a test set (1%). The validation set was used to identify the parameter α that controls the strength of the ℓ_2-regularization term. We considered discrete α-values of [0, 0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1.0]. The optimization procedure was implemented in PYTORCH [Paszke et al., 2019] and used the default LBFGS optimizer [Liu and Nocedal, 1989]. For the individual difference analyses, the procedure was identical except that we added a random effect for each participant and embedding dimension.

For the hold-out task analyses, the training set consisted of the concatenated choices13k and horizon task data. To obtain a validation and test set, we split the data of the experiential-symbolic task into eight folds and repeated the previously described fitting procedure for each fold. The validation set was used to identify the parameter α that controls the strength of the ℓ_2-regularization term and an inverse temperature parameter τ^{-1}. We considered discrete inverse temperature values of [0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1] and α-values as described above.

Model simulations: For the main analyses, we simulated model behavior by sampling from the predictions on the test set. For the hold-out task analyses, we simulated data deterministically based on a median threshold (again using the predictions on the test set). The resulting choice curves were generated by fitting a separate logistic regression model for each possible win probability of the E-option. Each model used the win probability of the S-option and an intercept term as independent variables and the probability of choosing the E-option as the dependent variable.

Baseline models: For the LLaMA baseline, we fitted an inverse temperature parameter to human choices using the procedure described above. For the BEAST baseline, we relied on the version provided for the choice prediction competition 2018 [Plonsky et al., 2018]. We additionally included an error model that selects a random choice with a particular probability. We treated this probability as a free parameter and fitted it using the procedure described above. The hybrid model closely followed the implementation of Gershman [2018]. We replaced the probit link function with a logit link function to ensure comparability to CENTaUR.

Supplementary Text

For the choices13k data set, we prompted each decision independently, thereby ignoring the potential effect of feedback. We used the following template:

> Machine 1 delivers 90 dollars with 10.0% chance and -12 dollars with 90.0% chance.
> Machine 2 delivers -13 dollars with 40.0% chance and 22 dollars with 60.0% chance.
>
> Your goal is to maximize the amount of received dollars.
>
> Q: Which machine do you choose?
> A: Machine [insert]

For the horizon task, we prompted each task independently, thereby ignoring potential learning effects across the experiment. We used the following template:

> You made the following observations in the past:
> - Machine 1 delivered 34 dollars.
> - Machine 1 delivered 41 dollars.

- Machine 2 delivered 57 dollars.
- Machine 1 delivered 37 dollars.

Your goal is to maximize the sum of received dollars within six additional choices.

Q: Which machine do you choose?
A: Machine [insert]

For the experiential-symbolic task, we prompted each decision independently and only considered the post-learning phase. We furthermore simplified the observation history by only including the option that is relevant to the current decision. We used the following template:

You made the following observations in the past:
- Machine 1 delivered 1 dollars.
- Machine 1 delivered 1 dollars.
- Machine 1 delivered -1 dollars.
- Machine 1 delivered 1 dollars.
- Machine 1 delivered 1 dollars.
- Machine 1 delivered -1 dollars.

[. . .]

- Machine 1 delivered 1 dollars.

Machine 2 delivers -1 dollars with 30.0% chance and 1 dollars with 70.0% chance.

Your goal is to maximize the amount of received dollars.

Q: Which machine do you choose?
A: Machine [insert]